

A Security Analysis of Anti-Audio Cloaking Schemes

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Abstract—Audio cloaking aims to protect audio artifacts from unauthorized cloning by speech synthesizers and to safeguard user identity by altering signals to make them unusable for voice synthesis and speaker identification. In this paper, we investigate the resilience of current audio cloaking methods from an adversarial perspective, analyzing whether an attacker can reverse these cloaking mechanisms to retrieve the original speaker features for accurate speaker verification and synthesis.

We systematically apply ten different audio restoration techniques, including adaptive frequency filtering, quantization-based filtering, ensemble averaging, and various spectral modifications, to cloaked audio samples. The effectiveness of each technique is rigorously evaluated using prominent speaker verification systems. Our findings show that most inversion attempts achieve limited success due to robust cloaking mechanisms; however, low-pass filtering emerged that is notable for its effectiveness, partially restoring speaker identity traits. WavLM achieved the highest similarity scores (up to 0.91) for low-pitch voices, and intelligibility was preserved with STOI scores greater than 0.28, partially reverse cloaking effects and restore speaker identity traits, especially in low-frequency voice profiles.

This research shows significant vulnerabilities in cloaking strategies, suggesting necessary improvements in future audio cloaking methods. By exposing these weaknesses, our study provides crucial information on enhancing the robustness of audio privacy and security measures.

Index Terms—Audio cloaking, adversarial inversion, speech synthesis, speaker verification.

I. INTRODUCTION

In today's world, the evolution of artificial intelligence and machine learning, along with the use of voice-based technologies, has become very important to modern digital experiences. Technologies such as virtual assistants like Siri and Alexa, which are used in secure authentication systems in banking and healthcare, rely on high-quality voice data to provide personalized and convenient services [1] [2] [3]. However, increasing reliance on voice data has also raised privacy and security concerns, as advanced audio synthesis and deep-fake technologies have made it easier for malicious actors to clone or manipulate voice recordings. This has caused a lot of damage, especially when voice samples are misappropriated for fraud, impersonation, or unauthorized access. [4] [5] [6] [7] [8]

For example, in 2023, scammers impersonated an elderly couple's grandson using AI-generated speech, nearly convincing them to send bail money before a bank manager intervened [9]. Similarly, in 2020, fraudsters cloned the voice of a company director to successfully request a \$35 million transfer from a bank manager, showing the severe financial risks posed by AI-generated voice cloning [10].

In response to this threat, researchers have developed audio cloaking techniques that alter the audio signal to prevent unauthorized use, particularly in tasks such as voice synthesis and speaker verification. These methods aim to maintain intelligibility while obscuring the unique vocal characteristics that synthesis models or speaker verification systems could exploit. The underlying principle of cloaking involves adding subtle perturbations or modifying the frequency components of the audio to make it unusable for synthesis without significantly affecting human perception [11] [12] [13]. Although audio cloaking shows promise as a privacy-enhancing technology, its effectiveness and resilience under adversarial conditions remain largely unexplored. An understanding of whether cloaked audio can be 'uncloaked' is essential to assess its reliability as a defense mechanism. By simulating attacks to reverse the cloaking process, this research seeks to evaluate the robustness of these techniques. Analyzing potential vulnerabilities will provide valuable information on the improvements needed to secure voice data against increasingly sophisticated threats.

A. Research Objectives

To examine the resilience of existing audio cloaking methods, this paper explores whether it is possible to reverse the cloaking process to reveal the original speaker's identity or vocal characteristics. Specifically, this study seeks to answer the following questions.

- 1) **Can cloak audio be restored to approximate the original speaker's voice?** By applying various audio restoration techniques, this research evaluates the extent to which speaker identity characteristics are preserved or obscured in cloaked audio.
- 2) **How effective are current speaker verification models in detecting restored cloaked audio?** This research tests the output of restoration techniques in speaker verification systems, to assess if the original identity can be verified.
- 3) **What are the specific vulnerabilities and limitations of current audio cloaking techniques?** This research identifies weaknesses in current cloaking methods and offers information on possible improvements to enhance their robustness against adversarial attacks.

By investigating these questions, we provide a comprehensive understanding of the resilience and limitations of existing audio cloaking techniques to protect against voice synthesis and speaker verification.

II. RELATED WORKS

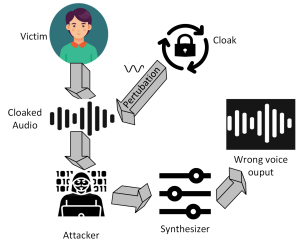


Fig. 1. Overview of Audio Cloaking

Speech synthesis attacks involve generating synthetic audio that closely mimics the voice of a target speaker, posing significant security risks by enabling impersonation. Although synthetic speech has existed since the 1930s [14], advances in deep neural networks (DNNs) have revolutionized this field, resulting in highly realistic voice cloning capabilities [15]–[17]. DNN-based models such as voice conversion (VC) and text-to-speech (TTS) allow attackers to replicate a speaker’s voice, raising concerns about identity theft and security vulnerabilities [18]–[20].

Voice Conversion systems modify the voice of a speaker to sound like a target, while Text To Speech systems synthesize speech from text using the voice profile of a target. These models can be used to produce arbitrary phrases or impersonations that threaten privacy and authenticity [21]. In response, research has explored both detection-based and cloaking-based defenses. Detection methods attempt to identify synthetic audio after generation using speaker verification (SV) and artifact detection models, but these approaches are susceptible to cross-language and recording quality degradation.

Cloaking-based techniques, on the other hand, proactively modify original speech signals prior to exposure to synthesis or verification systems. This prevents successful imitation or cloning. For example, *Protect Your Voice from Speech Synthesis Attacks* [12] introduces a frequency domain perturbation method that targets features used by synthesis models, while maintaining intelligibility for human listeners. Modifies specific spectral bands to reduce the fidelity of the cloning without affecting naturalness. Similarly, *AntiFake* [11] employs adversarial perturbations crafted to interfere with speaker embeddings used by TTS systems. Using ensemble learning, it ensures that the perturbations generalize across unknown black-box synthesizers, making it robust to transfer attacks.

In addition to these, *Real-Time Neural Voice Camouflage* [22] addresses the challenge of jamming ASR models during live audio streaming. It uses predictive adversarial attacks that preemptively introduce perturbations, maintaining effectiveness in real-time, over-the-air scenarios. Meanwhile, *HarmonyCloak* [23] adapts these ideas to music, embedding imperceptible noise to prevent generative AI models from learning and replicating compositions. These efforts collectively demonstrate that cloaking can be achieved through spectral, temporal, or embedding-space modifications that disrupt machine interpretability while maintaining human usability.

Despite these advances, recent work has highlighted the vulnerability of cloaked audio to inversion attacks. Attackers may apply signal processing or deep learning techniques to reverse cloaking, with the goal of restoring identifiable features. *Devil’s Whisper* [24] shows how adversarial examples targeting commercial ASR systems (such as Alexa and Siri) can be crafted even under black-box constraints using surrogate models and adaptive optimization. Similarly, studies such as [25] demonstrate that generative models can produce universal perturbations that are effective across many audio inputs, raising concerns about the generalizability of current cloaking defenses.

These inversion techniques often involve spectrum reconstruction, amplitude rescaling, or latent-space optimization using speaker verification feedback. The goal is to iteratively restore the original speaker embedding from a cloaked signal. Such attacks expose the limitations of current cloaking methods, particularly their reliance on fixed masking schemes or model-specific perturbations. They show the need for robust, adaptive, and transferable cloaking strategies that can withstand both white-box and black-box attacks.

Although audio cloaking offers a promising avenue for privacy-preserving voice communication, its resilience against adversarial attacks remains an open challenge. Our work contributes to this area by systematically evaluating the strengths and weaknesses of existing cloaking schemes under inversion attacks, guiding the development of more secure voice anonymization frameworks.

III. MODEL AND PROBLEM DEFINITION

We explore the feasibility of reversing the cloaking of audio from the attacker’s perspective. An adversary may only gain access to cloaked audio samples, without access to the original recordings, as illustrated in Figure 1. This scenario could occur when cloaked audio is transmitted over insecure channels or is publicly available online.

Given $X \in \mathbb{R}^T$ represents an audio signal of length T , containing speaker-identifying characteristics. The cloaking function $f : \mathbb{R}^T \rightarrow \mathbb{R}^T$ transforms X into a cloaked version $Y = f(X)$ such that:

$$SV(Y) \neq SV(X), \quad \text{and} \quad H(Y) \approx H(X)$$

where $SV(\cdot)$ is the speaker verification output (for example, an embedding), and $H(\cdot)$ is a human intelligibility function that ensures that the perturbed signal remains intelligible to listeners.

Following AntiFake [11], the cloaking process can be formally defined as an optimization problem that seeks a perturbation δ such that the output of a speaker encoder $g(\cdot)$ deviates from the original speaker identity, while remaining imperceptible to human perception:

$$\max_{\delta} \quad D(g(x_U + \delta), g(x_U)) - \alpha M(\delta)$$

subject to:

$$H(x_U) \approx H(x_U + \delta)$$

$$H(G(t, x_U)) \approx H(G(t, x_U + \delta))$$

$$SV(G(t, x_U)) \neq SV(G(t, x_U + \delta))$$

Here, x_U is the original user audio, $G(t, x)$ is a synthesis model that generates speech from text t and audio x , D is the distance in the speaker embedding space, $M(\delta)$ quantifies the magnitude of the disturbance and α is a tunable parameter that balances distortion and protection.

Suppose that an attacker intercepts $Y = f(X)$, the cloaked version of an original audio sample X , but does not have access to X itself. The attacker attempts to learn an approximate inverse transformation $g \approx f^{-1}$ such that the restored signal $\tilde{X} = g(Y)$ closely resembles the original speaker in identity.

This is evaluated using a speaker verification model $V(\cdot)$, which maps the audio to an embedding space. If the similarity between the embeddings $V(\tilde{X})$ and $V(X)$ exceeds a predefined threshold τ , the inversion is considered successful.

$$S(V(\tilde{X}), V(X)) > \tau$$

In practice, the attacker may use a sequence of transformations $g_1, g_2, \dots, g_n$, such as frequency filtering, quantization-dequantization, resampling, and denoising. The full inversion pipeline can be expressed as:

$$\tilde{X} = g_n \circ g_{n-1} \circ \dots \circ g_1(Y)$$

To formalize the attacker's goal, we define the anti-cloaking (inversion) optimization as:

$$\begin{aligned} & \max_{g \in \mathcal{G}} S(V(g(Y)), V(X)) \\ & \text{subject to } g = g_n \circ \dots \circ g_1, \quad S(V(g(Y)), V(X)) \geq \tau \end{aligned} \quad (1)$$

where $\mathcal{G}$ is the space of the allowed transformation pipelines and $S(\cdot, \cdot)$ is a similarity function (e.g. cosine similarity).

We evaluated the strength of the original cloaking function f by its ability to resist such inversion pipelines, that is, whether the similarity $S(V(\tilde{X}), V(X))$ remains below the threshold τ even after attempted restoration.

A. Speech Synthesis Models

To test inversion effectiveness, we use the SV2TTS model to determine whether restored audio allows successful cloning of the speaker's voice. SV2TTS synthesizes speech in three stages: a speaker encoder (LSTM), Tacotron 2-based spectrogram generation [26], and a WaveNet vocoder [27].

The speaker encoder maps 40-dimensional mel-spectrogram input to a 256-dimensional speaker embedding. We used the public implementation of SV2TTS, trained on VoxCeleb1/2 [28] and LibriSpeech [29] datasets. The default demo toolbox phrase from [30] was used for synthesis.

B. Speaker Verification Evaluation

We evaluated restored audio $\tilde{X}$ using three state-of-the-art speaker verification systems:

SpeechBrain: A cosine similarity threshold of 0.25 was used. Scores above this indicate successful restoration. PESQ MOS and STOI metrics were used to evaluate perceptual audio quality [31]–[33].

Resemblyzer: Known for its robustness to noise and artifacts, it used a threshold of 0.75. MOS was used to assess the similarity of the voices [34], [35].

WavLM: Using a similarity threshold of 0.86, WavLM offers strong identity discrimination. MOS scores were also reported [36].

This evaluation framework enables us to quantify both the strength of cloaking methods and the effectiveness of inversion techniques.

IV. ANTI-CLOAKING METHODOLOGIES

To reverse the cloaking effects on the target voice, we apply a variety of inversion techniques to the cloaked audio, then synthesize it, and compare its similarity to the original audio. Each technique was selected to test its potential to alter the cloaked audio.

A. Inversion Techniques

Each technique was carefully chosen to target specific types of cloaking transformations, from frequency adjustments to noise reduction. The following techniques were applied to each cloaked sample:

- 1) **Adaptive Frequency Filtering:** This reverses the effects of cloaking by dynamically applying high- and low-pass filters customized to the spectral characteristics of each audio sample [37]. The central idea is to enhance or suppress specific frequency bands based on the spectral centroid, which reflects where the energy of the signal is concentrated in the frequency domain.

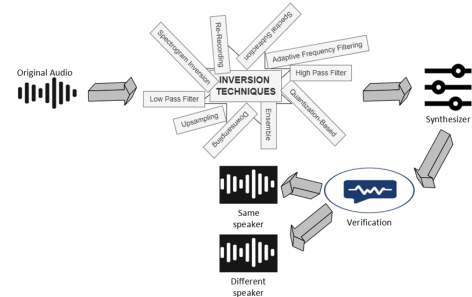


Fig. 2. inversion Techniques

The spectral centroid C , often interpreted as the "center of mass" of the spectrum, is calculated as follows:

$$C = \frac{\sum_{f=f_{HP}}^{f_{max}} f \cdot |Y(f)|}{\sum_{f=f_{HP}}^{f_{max}} |Y(f)|} \quad (2)$$

where f represents frequency bins, f_{max} is the maximum frequency of interest, and $|Y(f)|$ is the magnitude of the Fourier-transformed audio signal [37], [38].

Using this centroid, we define adaptive cutoff frequencies for the filters as:

$$f_{HP} = \alpha \cdot C, \quad f_{LP} = \beta \cdot C \quad (3)$$

where α and β are tunable scaling parameters that determine the sharpness and direction of filtering. This

formulation allows for dynamic targeting of spectral bands most likely affected by cloaking.

The processed signal $Y_{\text{filtered}}(t)$ is obtained by applying both filters to the input signal $Y(t)$:

$$Y_{\text{filtered}}(t) = H_{HP}(f_{HP}) \cdot Y(t) + H_{LP}(f_{LP}) \cdot Y(t) \quad (4)$$

where H_{HP} and H_{LP} denote the frequency responses of the high-pass and low-pass filters, respectively [39], [40]. We assume that cloaking modifies speaker-identifying frequency bands and that restoring spectral balance can recover speaker identity. This method is effective if cloaking disrupts predictable spectral structures but less effective when phase or non-linear distortions are introduced. The selection of appropriate values for α and β is crucial; poor tuning may cause spectral distortion or loss of important audio features. This approach also fails to address adversarial noise that is not related to spectral shaping.

- 2) Quantization Based: This approach takes advantage of the process of quantizing the audio signal at varying bit depths, such as 4-bit or 8-bit, followed by dequantization to smooth out fine details. The underlying idea is that quantization reduces the resolution of the signal, potentially suppressing cloaking artifacts embedded in subtle variations.

Formally, the quantization of an audio signal $Y(t)$, whose amplitude lies in the normalized range $[-1, 1]$, maps each sample to one of the discrete levels 2^B based on the specified bit depth B . The quantized signal is computed as:

$$Q_B(Y(t)) = \frac{\text{round}(Y(t) \cdot (2^B - 1))}{2^B - 1}$$

where round denotes the rounding to the nearest integer. After quantization, the signal is dequantized to return to the continuous domain:

$$Y_{\text{dequant}}(t) = Q_B(Y(t))$$

This dequantized signal suppresses fine-grained variations that may encode cloaking, potentially making the speaker's identity more recoverable [41]–[44].

We assume that cloaking artifacts appear in high-resolution signal details. Lowering bit depth can neutralize these artifacts through quantization, especially high-frequency adversarial noise. This method is simple, but it lacks precision for specific perturbations and may degrade speech quality if bit depth is too low. It is also ineffective against structural-level cloaking manipulations.

- 3) Ensemble Averaging of Multiple Samples: This technique creates slightly perturbed variants of a cloaked audio signal, using methods such as minor temporal shifts or additive noise, and averages them to produce a composite audio signal. The goal is to dilute the impact of individual perturbations introduced during cloaking, assuming that these distortions behave inconsistently under such modifications.

Given a cloaked signal $Y(t)$, we generate N modified versions $Y_i(t)$ for $i = 1, 2, \dots, N$, each incorporating slight changes. The final signal is formed through the following.

$$Y_{\text{avg}}(t) = \frac{1}{N} \sum_{i=1}^N Y_i(t)$$

This averaging process suppresses nonrepetitive perturbations, enhancing signal stability, and potentially restoring speaker identity.

Our assumption is that cloaking artifacts behave like stochastic noise and vary under minor input perturbations. By averaging out these inconsistencies, the underlying speaker characteristics may become clearer. We consider this technique robust against low-level randomized cloaking, but acknowledge that it is less effective when cloaking strategies introduce consistent deterministic transformations. It also requires additional computational cost as a result of generating and processing multiple samples.

- 4) Downsampling and Upsampling: This technique explores whether modifying the temporal resolution of cloaked audio through sample rate conversion can undo the effects of cloaking. We apply downsampling and upsampling independently to observe how the signal reconstruction changes and whether identity-related cues become more recoverable.

For downsampling, we reduce the sampling rate of the cloaked audio to values such as 4000 Hz, 8000 Hz, and 12000 Hz. This is done by retaining every D -th sample from the original signal:

$$y_{\text{down}}[n] = x[D \cdot n]$$

where D is the downsampling factor and $x[n]$ is the original signal. Before downsampling, a low-pass filter is applied to prevent aliasing, ensuring that high-frequency content above the new Nyquist frequency $\frac{\pi}{D}$ is removed. For upsampling, the sample rate is increased to 16000 Hz, 22050 Hz, or 24000 Hz by inserting $U - 1$ zeros between each sample:

$$y_{\text{up}}[n] = \begin{cases} y_{\text{orig}}\left(\frac{n}{U}\right), & \text{if } n = k \cdot U, k \in \mathbb{Z} \\ 0, & \text{otherwise} \end{cases}$$

A subsequent low-pass filter smooths out artifacts introduced during the zero-insertion process [45], [46].

This method assumes that the cloaking perturbations are sensitive to frequency resolution. By altering the temporal resolution, we may suppress or disrupt these perturbations. In our experience, this approach is computationally simple and can introduce meaningful distortions to the cloak. However, its effectiveness varies depending on how the cloaking mechanism was implemented. It is limited by the fact that significant downsampling can remove useful speaker characteristics and aggressive upsampling does not add new information, only interpolated guesses.

- 5) Mel-Spectrogram Inversion: This technique transforms the audio into a Mel spectrogram and then reconstructs

it using the Griffin-Lim algorithm. The intention is to regenerate a waveform that approximates the timbral structure of the original signal while potentially bypassing cloaking artifacts introduced in the time domain. To begin, the signal $x(t)$ is transformed using the Short-Time Fourier Transform (STFT) to generate a linear frequency spectrogram $S(f, t)$, which is then converted to the Mel scale:

$$M(f, t) = \text{MelScale}(|S(f, t)|)$$

Since the mel spectrogram lacks phase information, inversion is achieved using the Griffin-Lim algorithm. This iterative process reconstructs a phase estimate and refines it using the following steps.

- a) Initialize a phase matrix $\theta_0(f, t)$ with zeros or random values.
- b) Reconstruct an initial spectrogram: $S_0(f, t) = M(f, t)e^{j\theta_0(f, t)}$.
- c) At each iteration i :
 - Compute the inverse STFT to obtain $x_i(t)$.
 - Reapply STFT to get $S_i(f, t)$.
 - Update phase: $\theta_{i+1}(f, t) = \arg(S_i(f, t))$.
 - Update spectrogram: $S_{i+1}(f, t) = M(f, t)e^{j\theta_{i+1}(f, t)}$.
- d) Repeat until convergence is achieved.

Finally, the inverse STFT is applied to the final spectrogram to obtain $\hat{x}(t)$, the reconstructed waveform. We assume that converting to a Mel-based representation and reconstructing from it may suppress cloaking perturbations embedded in the waveform. This approach operates in a perceptually relevant domain and is effective for moderately cloaked signals. However, it fails when cloaking introduces severe phase alterations or irreversible frequency noise. Limitations include high computational cost and spectrogram inversion errors [47].

- 6) Adding a Second Cloak: This method investigates whether introducing an additional set of perturbations to an already cloaked signal can mitigate or unintentionally reverse the original cloaking effects. The idea stems from the concept of interference, where successive transformations may weaken or distort each other. The first cloaking operation can be expressed as:

$$x_{\text{cloaked}}(t) = x_{\text{original}}(t) + p_1(t)$$

where $p_1(t)$ denotes the initial perturbation applied to obscure speaker identity.

The second perturbation is then applied:

$$x_{\text{double_cloaked}}(t) = x_{\text{cloaked}}(t) + p_2(t)$$

Substituting the first equation into the second yields:

$$x_{\text{double_cloaked}}(t) = x_{\text{original}}(t) + p_1(t) + p_2(t)$$

The effectiveness of this approach hinges on the possibility that the new perturbation $p_2(t)$ might partially negate the effects of $p_1(t)$, restoring the acoustic features tied to the original speaker.

This method assumes that the cloak is not resilient to layered perturbations and that some destructive interference may result. It is an experimental and somewhat heuristic strategy. Our impression is that, while this approach lacks precision, it does provide insight into the fragility of some cloaking mechanisms. However, it may also exacerbate signal distortion, making intelligibility worse or rendering the signal further from the original identity.

- 7) Recording the Cloaked Audio: This method records cloaked audio using a phone setup in a room environment. The objective is to exploit analog channel distortions to corrupt or blur the digital perturbations introduced by the cloaking algorithm. Let $x_{\text{cloaked}}(t)$ represent the original cloaked signal. When recorded through a microphone, the resulting signal becomes

$$x_{\text{recorded}}(t) = x_{\text{cloaked}}(t) + \eta(t) + \delta(t)$$

where $\eta(t)$ represents system noise (e.g. thermal noise from the microphone) and $\delta(t)$ includes environmental effects such as echo, reverberation and ambient sound. The underlying assumption is that some perturbations in the cloak are fine enough to be lost or masked under realistic analogous imperfections. Although this technique is simple, our experience suggests that its utility lies in practical deployment scenarios rather than laboratory conditions. Its limitations include unpredictability and loss of control over signal fidelity due to uncontrolled environmental variables.

- 8) Spectral subtraction: Spectral subtraction is a classic noise reduction technique adapted here to identify and suppress frequency components likely added by the cloaking algorithm. It relies on estimating a noise profile and subtracting it from the original magnitude spectrum. The model assumes that the cloaked signal $y(k)$ is composed of the original audio $x(k)$ and a perturbation $n(k)$:

$$y(k) = x(k) + n(k)$$

From this we subtract the estimated noise spectrum $|N(e^{j\omega})|$ from the observed magnitude spectrum $|Y(e^{j\omega})|$ to estimate the clean signal:

$$|X(e^{j\omega})| = |Y(e^{j\omega})| - |N(e^{j\omega})|$$

Negative values are rectified to zero. The updated magnitude spectrum is used to reconstruct an approximation of the original signal.

This technique assumes that the perturbation resembles noise, particularly in low-energy speech segments. In our view, spectral subtraction is more effective against frequency-band cloaks than against adversarially crafted phase or time-domain distortions. It is limited by its dependence on accurate noise estimation and, when poorly tuned, it may suppress meaningful speech content.

- 9) High-Pass Filtering: This approach applies a filter that allows only frequencies above a specified cut-off f_c

to pass, removing low-frequency components where cloaking perturbations may reside.

Formally, the filtered signal is:

$$y(t) = x(t) * h_{HP}(t)$$

where $h_{HP}(t)$ is the high-pass filter kernel, and the frequency response is defined as:

$$H_{HP}(f) = \begin{cases} 1, & f > f_c \\ 0, & f \leq f_c \end{cases}$$

We optionally adjust the spectral centroid C to re-tune the filter behavior based on the energy distribution. The assumption is that the cloak might target low-frequency signals such as pitch and prosody. Our impression is that this method provides a blunt but effective way to emphasize discriminative features for speaker identification. However, it can also remove useful low-frequency information and introduce harsh artifacts.

- 10) Low-Pass Filtering with Multiplier Range Testing: In contrast to the previous method, this technique enhances the low-frequency band to test whether exaggerating these components disrupts the cloak or reveals suppressed patterns.

The low-pass filter passes frequencies below a cutoff f_c :

$$H_{LP}(f) = \begin{cases} 1, & f \leq f_c \\ 0, & f > f_c \end{cases}$$

We scale the output with a gain multiplier M ranging from 1.2 to 2.0:

$$y_{multiplied}(t) = M \cdot (x(t) * h_{LP}(t))$$

This technique assumes that cloaking mechanisms leave faint cues in low-frequency regions that become more prominent with amplification. We evaluated filters such as Butterworth (smooth), Chebyshev (sharp cut-off with ripple), and Bessel (phase-preserving). Although it has potential to uncover the masked energy, this method risks overwhelming the signal with low-frequency noise if M is too high or the filter choice is suboptimal.

V. PERFORMANCE EVALUATION

To assess the effectiveness of our anticloaking techniques, we evaluated speaker similarity and audio quality after inversion. Speaker embeddings were extracted using ECAPA-TDNN (SpeechBrain), Resemblyzer, and WavLM-Large. The cosine similarity between the original and inverted samples was used to measure identity preservation. For perceptual quality, we used neural network-based MOS estimators, PESQ MOS, and STOI intelligibility scores.

The performance of each technique was compared across these baselines and evaluated on its ability to recover the identity of the speaker and maintain intelligibility without introducing significant artifacts.

A. Dataset and Baselines

We performed our experiments using the ASVspoof2019 data set [48], selecting natural audio samples to evaluate the effectiveness of cloaking and anticloaking techniques. It served as a baseline, representing clean audio that was not affected by cloaking. To generate cloaked variants, we applied perturbations inspired by techniques proposed in *Protecting Your Voice from Speech Synthesis Attacks* and *AntiFake*, resulting in audio specifically designed to deceive speaker verification systems.

We first quantified the number of cloaked samples that successfully evaded recognition by three speaker verification models: SpeechBrain (ECAPA-TDNN), Resemblyzer, and WavLM. Table I summarizes how many samples were labeled as 'The Same Speaker' versus 'The Different Speaker' for both the AntiFake and Protect Your Audio datasets. These results established the baseline performance of speaker verification under cloaking and informed the effectiveness thresholds for our inversion techniques.

TABLE I
BASELINE COMPARISON OF SYNTHESIZED CLOAKED AUDIO

Dataset	Model	Same Speaker	Different Speaker	Total
AntiFake	SpeechBrain	46	454	500
	Resemblyzer	56	444	500
	WavLM	90	410	500
Protect Your Audio	SpeechBrain	400	100	500
	Resemblyzer	378	122	500
	WavLM	390	110	500

B. Experimental Setup

To evaluate the effectiveness of each anticloaking method, we implemented a controlled synthesis-verification pipeline. For every cloaked audio sample, we applied an inversion technique and then used the SV2TTS pipeline to synthesize speech from the processed audio. The synthesized output was then passed through three speaker verification models, SpeechBrain (ECAPA-TDNN), Resemblyzer, and WavLM, to assess whether the identity of the original speaker could be recovered.

All experiments were run on a machine equipped with an NVIDIA RTX 4070 Ti GPU using PyTorch and ONNX Runtime for accelerated inference. We ensured consistent preprocessing across samples, including normalization and resampling to 16 kHz.

Verification success was measured by whether the model classified the synthesized audio as the same speaker as the original uncloaked reference. Each inversion technique was evaluated for more than 500 cloaked samples per dataset variant (AntiFake and Protect Your Audio), and similarity scores were recorded alongside MOS, STOI and PESQ metrics to assess audio quality and intelligibility.

C. Experimental Results and Analysis

As shown in Figure 3, we evaluate various filtering and transformation techniques based on their attack success rate (ASR) defined as the percentage of samples that, after processing, are misclassified as the original speaker. A high

Fig. 3. Comparison of Various Audio Cloaking and Filtering Techniques Evaluated on AntiFake and Protect Your Voice Samples

Method	Metric	SpeechBrain		Resemblyzer		WavLM	
		AntiFake	Protect	AntiFake	Protect	AntiFake	Protect
Adaptive Frequency Filtering	Similarity Score	0.2703	0.1303	0.6443	0.6943	0.822	0.829
	Threshold Classification	Same Speaker	Diff. Speaker	Diff. Speaker	Diff. Speaker	Diff. Speaker	Diff. Speaker
	PESQ MOS Score	1.26	2.26	—	—	—	—
	MOS	—	—	1.20	1.30	1.26	1.40
	STOI Intelligibility	0.28	0.22	—	—	—	—
Quantization-Based Filtering	Similarity Score	0.19	0.27	0.70	0.69	0.84	0.81
	Threshold Classification	Diff. Speaker	Same Speaker	Diff. Speaker	Diff. Speaker	Diff. Speaker	Diff. Speaker
	PESQ MOS Score	1.09	2.26	—	—	—	—
	MOS	—	—	1.40	1.51	1.58	1.37
	STOI Intelligibility	0.32	0.19	—	—	—	—
Ensemble Averaging	Similarity Score	0.047	0.19	0.6802	0.6202	0.6686	0.8086
	Threshold Classification	Diff. Speaker	Diff. Speaker	Diff. Speaker	Diff. Speaker	Diff. Speaker	Diff. Speaker
	PESQ MOS Score	1.12	2.24	—	—	—	—
	MOS	—	—	1.05	1.85	1.23	2.23
	STOI Intelligibility	0.02	0.22	—	—	—	—
Downsampling (4k, 8k, 12kHz)	Similarity Score	0.133	0.294	0.67	0.66	0.85	0.93
	Threshold Classification	Diff. Speaker	Same Speaker	Diff. Speaker	Diff. Speaker	Same Speaker	Same Speaker
	PESQ MOS Score	1.19	2.40	—	—	—	—
	MOS	—	—	1.19	1.03	1.37	1.43
	STOI Intelligibility	0.31	0.25	—	—	—	—
Upsampling (16k, 22k, 24kHz)	Similarity Score	0.085	0.265	0.74	0.68	0.85	0.93
	Threshold Classification	Diff. Speaker	Same Speaker	Diff. Speaker	Diff. Speaker	Same Speaker	Same Speaker
	PESQ MOS Score	1.16	1.82	—	—	—	—
	MOS	—	—	1.02	1.32	1.29	1.27
	STOI Intelligibility	0.28	0.17	—	—	—	—
Mel-Spectrogram Inversion	Similarity Score	0.20	0.26	0.7185	0.6578	0.82	0.8130
	Threshold Classification	Diff. Speaker	Same. Speaker	Diff. Speaker	Diff. Speaker	Diff. Speaker	Diff. Speaker
	PESQ MOS Score	1.82	2.08	—	—	—	—
	MOS	—	—	1.33	1.52	1.12	1.32
	STOI Intelligibility	0.08	0.20	—	—	—	—
Second Cloak Application	Similarity Score	0.132	0.14	0.73	0.59	0.82	0.726
	Threshold Classification	Diff. Speaker	Diff. Speaker	Diff. Speaker	Diff. Speaker	Diff. Speaker	Diff. Speaker
	PESQ MOS Score	1.18	2.19	—	—	—	—
	MOS	—	—	1.33	1.43	1.16	1.2
	STOI Intelligibility	0.23	0.22	—	—	—	—
Simulated + Re-Recording	Similarity Score	0.06	0.21	0.69	0.649	0.80	0.84
	Threshold Classification	Diff. Speaker	Diff. Speaker	Diff. Speaker	Diff. Speaker	Diff. Speaker	Diff. Speaker
	PESQ MOS Score	1.15	2.25	—	—	—	—
	MOS	—	—	0.98	1.14	1.24	1.04
	STOI Intelligibility	0.26	0.18	—	—	—	—
Spectral Subtraction	Similarity Score	0.11	0.26	0.75	0.71	0.82	0.82
	Threshold Classification	Diff. Speaker	Same Speaker	Same Speaker	Diff. Speaker	Diff. Speaker	Diff. Speaker
	PESQ MOS Score	1.17	2.16	—	—	—	—
	MOS	—	—	1.27	1.07	1.02	1.21
	STOI Intelligibility	0.26	0.15	—	—	—	—
High Pass Filtering	Similarity Score	0.21	0.11	0.6501	0.5801	0.87	0.71
	Threshold Classification	Diff. Speaker	Diff. Speaker	Diff. Speaker	Diff. Speaker	Same Speaker	Diff. Speaker
	PESQ MOS Score	1.61	2.36	—	—	—	—
	MOS	—	—	1.07	2.27	1.19	1.92
	STOI Intelligibility	0.06	0.19	—	—	—	—
Low Pass Filtering	Similarity Score	0.34	0.29	0.84	0.75	0.91	0.85
	Threshold Classification	Same Speaker	Same Speaker	Same Speaker	Same Speaker	Same Speaker	Same Speaker
	PESQ MOS Score	1.06	2.55	—	—	—	—
	MOS	—	—	1.38	1.85	1.49	1.90
	STOI Intelligibility	0.28	0.21	—	—	—	—

ASR (that is, the “Same Speaker” classification with high similarity) indicates failure of the cloaking protection, while a low ASR (that is, the “Different Speaker” classification with low similarity) reflects successful cloaking.

Each technique was applied to cloaked samples, followed by synthesis and verification. The results reflect the average performance of 500 samples per method. The following summary presents effectiveness trends.

Adaptive Frequency Filtering yielded low ASR across all models except SpeechBrain-Protect, indicating limited success in altering speaker identity via spectral rebalancing. Low

PESQ and STOI scores also indicate acoustic degradation, which can help hinder inversion.

Quantization-Based Inversion also produced low ASR across Resemblyzer and WavLM, although SpeechBrain misclassified some Protect samples as the original speaker, suggesting that reducing signal resolution does not sufficiently modify speaker features to reverse the cloak.

Ensemble Averaging showed limited impact. Minor perturbations averaged across the samples failed to significantly disrupt the cloaking pattern.

Downsampling at 4000 to 12000 Hz and upsampling at

16000 to 24000 Hz produced notable increases in ASR for certain models, particularly WavLM and SpeechBrain-Protect, indicating that frequency resolution manipulation can partially reverse cloaking. However, the effect was inconsistent across models, as Resemblyzer maintained a low ASR, suggesting that frequency-based transformations alone are insufficient for universal inversion.

The Mel Spectrogram Inversion showed low ASR as well. Despite spectral approximation through Griffin-Lim, phase loss likely hindered full identity restoration.

The second application of the cloak (double perturbation) showed poor results, affirming that overlaying cloaks does not reverse identity masking.

Simulated + Re-Recording, which introduces environmental artifacts, preserved the cloaked identity with low similarity and the results of 'Different Speaker' across all models. This confirms the strength of the cloaking even under signal distortion.

Spectral subtraction produced inconsistent results across models. Although SpeechBrain-Protect and WavLM-Protect produced high similarity scores (0.26 and 0.82 respectively) and were classified as 'The same speaker', other combinations of model samples maintained low similarity and 'Different Speaker' classification.

High-pass filtering has limited effectiveness in reversing cloaking. Only WavLM-AntiFake was classified as the 'Same Speaker' with a high similarity score of 0.87, indicating partial identity restoration. However, SpeechBrain and Resemblyzer remained largely unaffected, indicating that high-frequency emphasis alone is not reliably effective.

In contrast, Low-Pass Filtering was the only technique that consistently reversed the cloak across all models. Every sample of AntiFake and Protect was classified as 'Same Speaker', with very high similarity scores (e.g. 0.84 to 0.90 for WavLM) and supporting metrics like MOS and STOI showed minimal distortion. This shows that focusing on low-frequency features successfully restores speaker-specific characteristics that were masked by the original cloak. In summary, as shown in Figure 4, low-pass filtering was the most effective method to break cloaking protections in all models. In contrast, techniques such as downsampling, upsampling, spectral subtraction, and high-pass filtering demonstrated only limited or model-dependent effectiveness.

Further Study: Effectiveness of Low-Pass Filtering for Cloaking Inversion

To assess the potential of low-pass filtering in reversing cloaked audio, we applied filters with multipliers ranging from 1.5 to 2.0 and evaluated their impact on speaker verification outcomes using the *AntiFake* and *Protect Your Audio* datasets. Table II summarizes the number of samples classified as the 'Same Speaker' by three models after synthesis. The observations from these results suggest that for the *AntiFake* dataset, the application of low-pass filtering led to a marked increase in the number of "Same Speaker" classifications, especially in WavLM, which saw a shift from 25 to 67 classifications at multiplier 1.9. Resemblyzer and SpeechBrain also showed modest improvements in the multiplier range. These trends indicate that *AntiFake*'s cloaking mechanism is

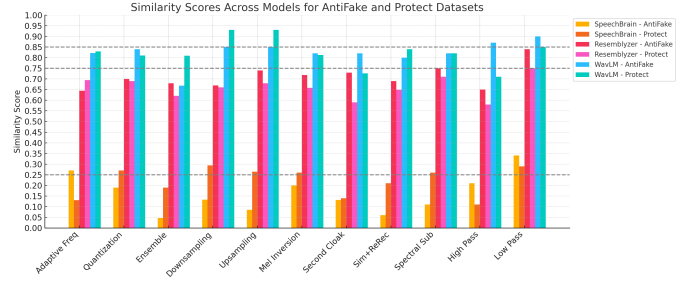


Fig. 4.

more susceptible to frequency-domain restoration strategies, especially in the lower frequency spectrum targeted by low-pass filters. In contrast, for the *Protect Your Audio* dataset, the recognition of "Same Speaker" remained high across all multipliers, with values starting from 55 and peaking at 80 for SpeechBrain. This raises a critical observation; since the baseline speaker identity remained partially intact even before filtering, the improvements from low-pass filtering may not necessarily reflect successful inversion. Rather, it is plausible that *Protect Your Audio* did not fully obfuscate speaker-specific characteristics, making the apparent gains a result of reinforcement rather than reversal. Low-pass filtering demonstrated tangible potential as an inversion technique, especially to undo the effects of more aggressive cloaks such as those used in *AntiFake*. However, the interpretation of results for *Protect Your Audio* remains nuanced and warrants further investigation to distinguish between genuine restoration of speaker identity and amplification of residual speaker traits.

TABLE II
LOW-PASS FILTERING RESULTS FOR ANTIFAKE AND PROTECT YOUR AUDIO DATASETS

Multiplier	Model	Same (AntiFake)	Different (AntiFake)	Same (Protect)	Different (Protect)	Total
1.5	SpeechBrain	31	69	55	45	100
	Resemblyzer	25	75	59	41	100
	WavLM	25	75	58	42	100
1.6	SpeechBrain	23	77	60	40	100
	Resemblyzer	27	73	57	43	100
	WavLM	27	73	50	50	100
1.7	SpeechBrain	15	85	63	37	100
	Resemblyzer	19	81	62	38	100
	WavLM	47	53	67	33	100
1.8	SpeechBrain	65	35	57	43	100
	Resemblyzer	50	50	50	50	100
	WavLM	45	55	62	38	100
1.9	SpeechBrain	60	40	75	25	100
	Resemblyzer	62	38	70	30	100
	WavLM	67	33	72	28	100
2.0	SpeechBrain	40	60	80	20	100
	Resemblyzer	34	66	72	28	100
	WavLM	55	45	76	24	100

TABLE III
COMPARISON OF HIGH AND LOW PITCH VOICE PERFORMANCE ACROSS MODELS

Metric	SpeechBrain		Resemblyzer		WavLM	
	High	Low	High	Low	High	Low
Similarity Score	0.25	0.40	0.79	0.88	0.86	0.91
Threshold Classification	Same	Same	Same	Same	Same	Same
PESQ MOS Score	1.06	1.23	—	—	—	—
MOS	—	—	1.10	2.02	1.02	1.98
STOI Intelligibility	0.06	0.28	—	—	—	—

To assess the impact of low-pass filtering on cloaked audio, we analyzed its effect on high-pitch and low-pitch voices. High-pitch voices, rich in high-frequency energy, are more affected by filtering, which attenuates crucial spectral details.

SpeechBrain performed poorly on high-pitch samples (similarity score: 0.25, PESQ MOS: 1.06), while Resemblyzer (0.79, MOS: 1.10) and WavLM (0.86, MOS: 1.02) preserved more speaker identity but with reduced perceptual quality. In contrast, low-pitch voices, with more energy in preserved low-frequency bands, showed better results across all models. SpeechBrain improved to 0.40 similarity, and both Resemblyzer and WavLM achieved 0.88 and 0.91, respectively. These results, summarized in Table III, confirm that low-pass filtering is more effective for low-pitch voices. Although it can partially restore high-pitch voices, this often comes at the cost of intelligibility and quality. The low-pass filtering between 150% - 200% shows varying effectiveness based on the pitch and model of the speaker. SpeechBrain performed poorly on high-pitch voices, but improved for low-pitch ones. Resemblyzer and WavLM were consistent across both, with slightly better results for low-pitch voices. Overall, low-pass filtering partially reverses cloaking, especially for low-pitch profiles, though model performance varies.

VI. DISCUSSION AND FUTURE WORK

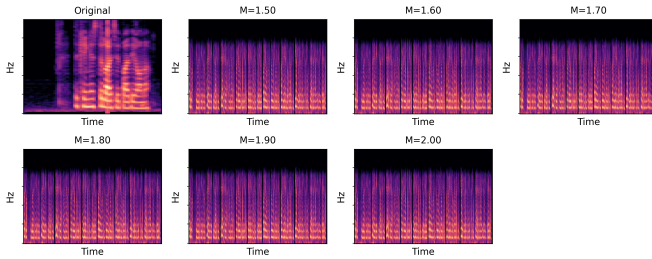


Fig. 5. Mel spectrograms of original and low-pass filtered audio (150% multiplier).

Only low-pass filtering showed partial success, particularly with spectral centroid multipliers between 150% and 200%. Spectrograms (Figure 5) confirm that low-pass filtering attenuates high-frequency energy while preserving lower-frequency content. This preserves intelligibility but reduces speaker-specific timbre, limiting full identity recovery. The performance of the model varied by pitch. SpeechBrain performed better on low pitch voices, with a similarity score of 0.40 compared to 0.25 for high pitch voices. The assembly machine and WavLM maintained higher similarity scores across both pitch ranges but still favored low-pitch input. These results suggest that low-pass filtering is more effective for low-pitch voices, where key identity features lie in preserved frequency bands. AntiFake samples, which used ensemble perturbations, were more resistant to inversion. Even after filtering, artifacts remained due to dispersed perturbations across time and frequency. Future research should explore adversarially robust cloaking, develop learning-based inversion methods (with ethical safeguards), and improve verification models that can detect cloaked or restored speech. A more comprehensive evaluation is also needed for accents, genders, and age groups. Although current cloaking techniques remain effective, they are not immune to targeted restoration. As

synthesis and inversion methods improve, cloaking will require continuous adaptation to preserve speaker privacy.

VII. LIMITATION

Our research faced limitations including reduced perceptual quality resulting from inversion techniques, significant dependence on pitch and speaker verification models, a limited exploration of complex phase or temporal cloaks, and restricted generalizability due to dataset diversity.

VIII. CONCLUSION

Our evaluation of audio cloaking techniques using multiple restoration methods highlighted low-pass filtering as particularly effective in countering frequency-based cloaking, especially for low-pitch voices. The findings emphasize frequency-domain vulnerabilities and the influence of pitch on inversion success, underscoring the need for continuous adaptation and improvement of cloaking techniques to ensure robust audio privacy.

REFERENCES

- [1] Wikipedia Contributors. (2024) Speaker recognition. Accessed: 2024-11-08. [Online]. Available: https://en.wikipedia.org/wiki/Speaker_recognition
- [2] SkillUpwards. (2024) Ai technologies behind voice assistants like siri and alexa. Accessed: 2024-11-08. [Online]. Available: <https://skillupwards.com/blog/ai-technologies-behind-voice-assistants-like-siri-and-alexa>
- [3] EnterpriseBot AI. (2024) Ai voice assistants in healthcare: Top use cases. Accessed: 2024-11-08. [Online]. Available: <https://www.enterprisebot.ai/blog/ai-voice-assistants-in-healthcare-top-use-cases>
- [4] Security Intelligence. (2024) A new wave of deepfake cybercrime. Accessed: 2024-11-08. [Online]. Available: <https://securityintelligence.com/posts/new-wave-deepfake-cybercrime/>
- [5] Wired. (2024) How to protect yourself from ai scam calls. Accessed: 2024-11-08. [Online]. Available: https://www.wired.com/story/how-to-protect-yourself-ai-scam-calls-detect?utm_source=chatgpt.com
- [6] American Bankers Association. (2024) Challenges in voice biometrics: Vulnerabilities in the age of deepfakes. Accessed: 2024-11-08. [Online]. Available: <https://bankingjournal.aba.com/2024/02/challenges-in-voice-biometrics-vulnerabilities-in-the-age-of-deepfakes/>
- [7] Resemble AI. (2024) Deepfake voice generators and data privacy. Accessed: 2024-11-08. [Online]. Available: <https://www.resemble.ai/deepfake-voice-generators-data-privacy/>
- [8] Cloud Security Alliance. (2024) Ai and deepfake security concerns. Accessed: 2024-11-08. [Online]. Available: <https://cloudsecurityalliance.org/blog/2024/06/25/ai-deepfake-security-concerns/>
- [9] T. W. Post. (2023) They thought loved ones were calling for help. it was an ai scam. The Washington Post. Accessed: 2024-11-09. [Online]. Available: <https://www.washingtonpost.com/technology/2023/ai-scam-voice-cloning-family/>
- [10] I-AML. (2020) Uae police: Fraudsters cloned company director's voice in \$35 million bank heist. I-AML. Accessed: 2024-11-09. [Online]. Available: <https://i-aml.com/uae-police-fraudsters-cloned-company-directors-voice-in-35-million-bank-heist/>
- [11] Z. Yu, S. Zhai, and N. Zhang, "Antifake: Using adversarial audio to prevent unauthorized speech synthesis," in *Proceedings of the 2023 ACM SIGSAC Conference on Computer and Communications Security*, 2023, pp. 460–474.
- [12] Z. Liu, Y. Zhang, and C. Miao, "Protecting your voice from speech synthesis attacks," 12 2023, pp. 394–408.
- [13] Deng, Jiangyi and others, "V-Cloak: Real-Time Voice Anonymization System," in *Proceedings of the 2023 USENIX Security Symposium*, 2023, accessed: 2024-11-08. [Online]. Available: <https://www.usenix.org/system/files/usenixsecurity23-deng-jiangyi-v-cloak.pdf>
- [14] J. Mullennix and S. Stern, *Computer Synthesized Speech Technologies: Tools for Aiding Impairment*. IGI Global, 2010.

- [15] G. K. Anumanchipalli, K. Prahallad, and A. W. Black, "Festvox: Tools for creation and analyses of large speech corpora," in *Workshop on Very Large Scale Phonetics Research, UPenn, Philadelphia*, vol. 70, 2011.
- [16] P. Lanchantin, G. Degottex, and X. Rodet, "A HMM-based speech synthesis system using a new glottal source and vocal-tract separation method," in *2010 IEEE International Conference on Acoustics, Speech and Signal Processing*. IEEE, 2010, pp. 4630–4633.
- [17] K. Tokuda, Y. Nankaku, T. Toda, H. Zen, J. Yamagishi, and K. Oura, "Speech synthesis based on hidden Markov models," *Proceedings of the IEEE*, vol. 101, no. 5, pp. 1234–1252, 2013.
- [18] E. Casanova, J. Weber, C. D. Shulby, A. C. Junior, E. Gölge, and M. A. Ponti, "Yourtts: Towards zero-shot multi-speaker TTS and zero-shot voice conversion for everyone," in *International Conference on Machine Learning*. PMLR, 2022, pp. 2709–2720.
- [19] Y. Jia, Y. Zhang, R. Weiss, Q. Wang, J. Shen, F. Ren, P. Nguyen, R. Pang, I. L. Moreno, Y. Wu *et al.*, "Transfer learning from speaker verification to multispeaker text-to-speech synthesis," in *Advances in Neural Information Processing Systems*, vol. 31, 2018.
- [20] H. Kameoka, T. Kaneko, K. Tanaka, Y. Y. Lin, C.-M. Chien, J.-H. Lin, H. yi Lee, and L. shan Lee, "Fragmentvc: Any-to-any voice conversion by end-to-end extracting and fusing fine-grained voice fragments with attention," in *ICASSP 2021-2021 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2021, pp. 5939–5943.
- [21] L. M. Onyekwelu-Udoka, "AI-generated Synthetic Audio Analysis and Forensic Attribution," dec 2024.
- [22] M. Chiquier, C. Mao, and C. Vondrick, "Real-time neural voice camouflage," in *ICLR*, 2022. [Online]. Available: <https://arxiv.org/abs/2112.07076>
- [23] S. I. A. Meerza, L. Sun, and J. Liu, "Harmonycloak: Making music unlearnable for generative ai," *University of Tennessee, Knoxville*, 2024. [Online]. Available: <https://mosis.eecs.utk.edu/publications/meerza2024harmonycloak.pdf>
- [24] Y. e. a. Chen, "Devil's whisper: A general approach for physical adversarial attacks against commercial black-box speech recognition devices," in *USENIX Security*, 2020. [Online]. Available: https://kaichen.org/paper/conference/sec20summer_chen-yuxuan_prepub.pdf
- [25] Y. Xie, Z. Li, C. Shi, J. Liu, Y. Chen, and B. Yuan, "Enabling fast and universal audio adversarial attack using generative model," *arXiv preprint arXiv:2004.12261*, 2020.
- [26] Y. Wang, R. Skerry-Ryan, D. Stanton, Y. Wu, R. J. Weiss, N. Jaitly, Z. Yang, Y. Xiao, Z. Chen, S. Bengio *et al.*, "Tacotron: Towards end-to-end speech synthesis," *arXiv preprint arXiv:1703.10135*, 2017.
- [27] A. v. d. Oord, S. Dieleman, H. Zen, K. Simonyan, O. Vinyals, A. Graves, N. Kalchbrenner, A. Senior, and K. Kavukcuoglu, "Wavenet: A generative model for raw audio," *arXiv preprint arXiv:1609.03499*, 2016.
- [28] A. Nagrani, J. S. Chung, and A. Zisserman, "Voxceleb: a large-scale speaker identification dataset," *arXiv preprint arXiv:1706.08612*, 2017.
- [29] V. Panayotov, G. Chen, D. Povey, and S. Khudanpur, "Librispeech: an asr corpus based on public domain audio books," in *2015 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2015, pp. 5206–5210.
- [30] Corentin Jemine, "Real-Time Voice Cloning," 2024, accessed: 2024-11-08. [Online]. Available: <https://github.com/CorentinJ/Real-Time-Voice-Cloning>
- [31] SpeechBrain Contributors, "Speechbrain: Open-source toolkit for speech processing," 2024, cosine similarity threshold set at 0.25, indicating obfuscation for scores below this value. [Online]. Available: <https://speechbrain.github.io/documentation>
- [32] A. Rix, J. Beerends, M. Hollier, and A. Hekstra, "Perceptual evaluation of speech quality (pesq) - a new method for speech quality assessment of telephone networks and codecs," in *Proceedings of the 2001 IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*, vol. 2, 2001, pp. 749–752. [Online]. Available: <https://ieeexplore.ieee.org/document/941023>
- [33] C. Taal, R. Hendriks, R. Heusdens, and J. Jensen, "An algorithm for intelligibility prediction of time-frequency weighted noisy speech," *IEEE Transactions on Audio, Speech, and Language Processing*, vol. 19, no. 7, pp. 2125–2136, 2011. [Online]. Available: <https://doi.org/10.1109/TASL.2011.2114881>
- [34] Resemblyzer Contributors, "Resemblyzer: Open-source voice embedding framework," 2024, threshold set at 0.75, where lower scores show successful obfuscation. [Online]. Available: <https://github.com/resemble-ai/Resemblyzer>
- [35] Wikipedia Contributors. (2024) Mean opinion score. Accessed: 2024-11-08. [Online]. Available: https://en.wikipedia.org/wiki/Mean_opinion_score
- [36] S. Chen, C. Wang, Z. Chen, Y. Wu, S. Liu, Z. Chen, J. Li, N. Kanda, T. Yoshioka, X. Xiao, J. Wu, L. Zhou, S. Ren, Y. Qian, Y. Qian, J. Wu, M. Zeng, and F. Wei, "Wavlm: Large-scale self-supervised pre-training for full stack speech processing," 2021.
- [37] K. T. Andersen and M. Moonen, "Adaptive time-frequency analysis for noise reduction in an audio filter bank with low delay," *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 24, no. 4, pp. 784–795, 2016. [Online]. Available: <https://ieeexplore.ieee.org/document/7400994>
- [38] G. Peeters, B. Giordano, P. Susini, N. Misdariis, and S. McAdams, "The timbre toolbox: Extracting audio descriptors from musical signals," *The Journal of the Acoustical Society of America*, vol. 130, no. 5, pp. 2902–2916, 2011.
- [39] DSP Stack Exchange Community, "Calculating the spectral centroid of a signal," <https://dsp.stackexchange.com/questions/38773/calculating-the-spectral-centroid-of-a-signal>, 2017, accessed: YYYY-MM-DD.
- [40] J. O. Smith, *Spectral Audio Signal Processing*. W3K Publishing, 2011.
- [41] C. E. Shannon, "A mathematical theory of communication," *The Bell System Technical Journal*, vol. 27, no. 3, pp. 379–423, 623–656, 1948. [Online]. Available: <https://ieeexplore.ieee.org/document/6773024>
- [42] H. Nyquist, "Certain topics in telegraph transmission theory," *Transactions of the American Institute of Electrical Engineers*, vol. 47, no. 2, pp. 617–644, 1928. [Online]. Available: <https://ieeexplore.ieee.org/document/5055024>
- [43] J. G. Proakis and D. G. Manolakis, *Digital Signal Processing: Principles, Algorithms, and Applications*, 4th ed. Prentice Hall, 2006.
- [44] A. V. Oppenheim and R. W. Schaffer, *Discrete-Time Signal Processing*, 3rd ed. Pearson, 2009.
- [45] R. E. Crochiere and L. R. Rabiner, *Multirate Digital Signal Processing*. Englewood Cliffs, NJ: Prentice-Hall, 1983.
- [46] J. O. I. Smith, *Multirate Signal Processing for Communication Systems*, 2020. [Online]. Available: https://ccrma.stanford.edu/~jos/JFB/JFB_2up.pdf
- [47] D. W. Griffin and J. S. Lim, "Signal estimation from modified short-time fourier transform," *IEEE Transactions on Acoustics, Speech, and Signal Processing*, vol. 32, no. 2, pp. 236–243, 1984. [Online]. Available: <https://ieeexplore.ieee.org/document/1164317>
- [48] M. Todisco, A. Nautsch, N. Evans, T. Kinnunen, J. Yamagishi, A. Kanervisto, and M. Sahidullah, "Asvspoof 2019: Automatic speaker verification spoofing and countermeasures challenge evaluation plan," in *Proc. Interspeech 2019*, 2019, pp. 1013–1017.